Embedded Spaces for Discovering New Classes

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Abstract

1 Introduction

A representation can encode a novel class perfectly and still be useless for finding it. This is not a hypothetical failure mode; it is the ordinary situation. Train a supervised contrastive encoder on a set of known classes, hold one class out, and a linear probe trained to separate the held-out class from the rest reaches an AUC of 0.94. The same space, asked to flag novel examples without being told what to look for, scores exactly zero at the event level. The information is present and linearly accessible. It is simply not accessible to anyone who does not already have the answer.

The gap between those two numbers is the subject of this paper. It is the gap between a representation being *decodable* and being *discoverable*, and it matters because scientific discovery lives entirely on the second side of it. A search for a new particle, a new galaxy morphology, or a new failure mode of a deployed model does not come with labels for the thing being searched for. What it has is a large corpus, a smaller reference sample of what is already understood, and a question: is there something here that the reference does not explain?

Most of the representation-learning literature optimises for the first side. Linear probe accuracy, k-NN accuracy, and transfer benchmarks all measure how well a space supports a decision that someone else has already specified. They are the right metrics for that purpose. They are close to uninformative about whether a space will surface structure nobody has named yet, and — as the two numbers above indicate — they can be actively misleading, because the training objectives that maximise them are not the objectives that make distances mean anything in absolute terms.

What we do. We treat discovery as the target and work backwards. The pipeline we study is the standard open-world loop: score every point in a corpus against a reference of labelled examples, take the most anomalous ones as a *pool*, cluster the pool to propose new class anchors, and adapt the representation to the proposal. Every stage of that loop is downstream of one quantity — the fraction of the pool that is genuinely novel, which we call *pool purity*. Purity is the only forward-looking metric in the pipeline. Every other measure describes a space you already have; purity predicts whether the next step will improve it or corrupt it. Below a threshold the loop does not merely fail to help: it clusters background, plants anchors on it, and fine-tunes the representation to make that structure sharper.

Contributions.

1. **An algebra for the pool, and a ceiling.** Writing b for the novel base rate, q for the pool fraction, and ε_s for the fraction of novel points retained, pool purity is $\Pi = \varepsilon_s b/q$, so the enrichment Π/b is exactly ε_s/q and $\Pi \leq \min(1, b/q)$. The ceiling is a property of the cut alone. It says that a pool threshold inherited without derivation — as ours was — can cap achievable purity far below the gate no matter how good the representation is, and it separates "the space is bad" from "the cut is bad" for the first time.

2. **A label-free rule for the cut.** Because purity requires the novel labels, tuning the cut on purity is oracle tuning. We derive the novel-count estimator that the Neyman–Pearson critic

already provides, and select the tightest cut that retains enough *estimated* novel points to remain clusterable. The rule uses only the corpus and the labelled reference. On synthetic data with an exact critic it recovers the oracle cut exactly; on real embeddings it raises pool purity from 0.232 to 0.778 where the scorer is strong, and refuses — reporting that it has refused — where it is not.

3. **A regime split, stated as a condition rather than a caveat.** Single-class novelty at a rate of $\sim$1% and multi-class novelty at $\sim$10% are different problems with different answers: which pooling statistic wins, whether stricter thresholds help or hurt, and whether the gate is reachable at all invert between them. We report them separately throughout.

4. **A reduced set of objectives.** From a design cube of ten contrastive objectives we identify the few that earn a place for discovery specifically, and show that the ranking by discovery is not the ranking by probe. On a from-scratch CIFAR-100 lineage, adding sliced-isotropy to a supervised contrastive objective is a tie on the linear probe (0.937 vs. 0.940) and costs 1.1 points of top-1, while raising every detection column — Euclidean AUC $0.233 \to 0.620$, tied-Mahalanobis $0.279 \to 0.602$. One point of accuracy buys a $2.7\times$ improvement in the geometry discovery actually runs on.

5. **Two constructions, and a decomposition of what they buy.** We study a residual decomposition — train a second encoder on what the anchors fail to explain, then concatenate — and frozen-space discovery, in which only the anchors move. Because both can be composed with the discovery loop, we report them in a 2×2 ablation (construction $\times$ discovery) rather than as a single delta, which separates a genuine interaction from an additive stack. The composition is superadditive in 12 of 20 cells but concentrated: the effect that takes per-event detection from nominal to $6\times$ nominal lives in one cell, while the two largest gains elsewhere turn out to be main effects of the construction and of discovery separately (Section 4). We state this narrowly because the decomposition is what makes it checkable.

6. **A leakage-free lineage for the claims that need one.** Every CIFAR cell built from hub-pretrained weights inherits an encoder that already saw the held-out class, which makes it unusable as evidence for discovery. We rebuild the CIFAR battery from random initialisation and report the pretrained cells only as an ablation. Dataset ordering throughout is by distance from pretraining rather than alphabetically, so the reader can see which claims rest on a backbone that could have seen the answer.

What this paper is not. We do not claim that a regulariser which makes embedded distance behave like a log-likelihood thereby improves discovery; we tested that directly and it does not hold (Section 6). We do not claim the pipeline works everywhere. A substantial part of what follows is a characterisation of where it does not, and why — in several cases the obstruction is the novel-class rate rather than anything about the representation, which is a limit no encoder can move.

2 Setup: discovery as a two-sample problem

The open-world protocol. We are given a corpus $\mathcal{D}$ of N embedded points, of which an unknown fraction b belong to classes never seen in training, and a reference $\mathcal{R} \subset \mathcal{D}$ of points carrying known class labels. No label for any novel class is available at any point — not for choosing hyperparameters, not for setting thresholds. We state this explicitly because the constraint is easy to violate accidentally, and because a method that quietly violates it reports numbers that cannot be reproduced in the situation it claims to address.

The loop. One round proceeds as follows. Score every point by an anomaly score $s(x)$; threshold to obtain a pool $\mathcal{P}$; cluster $\mathcal{P}$ and append the resulting centroids to the anchor set; pseudo-label every pooled point by its nearest discovered anchor; and adapt the representation (or, in the frozen variant of Section 4, only the anchors) to the enlarged anchor set. The round repeats with the updated anchors.

Pool purity, and why it gates everything. Let Π denote the fraction of $\mathcal{P}$ that is truly novel. The pool feeds directly into a clustering step with no supervision downstream of it, so $\mathcal{P}$ is the last point at which a mistake can still be corrected and nothing after it checks the work. At $\Pi \approx 0.01$ the clustering is not finding the novel class; it is partitioning background, and the subsequent fine-tune sharpens that partition. Empirically the probe drops by 0.031 to 0.038 wherever such pools form. Purity therefore behaves as a gate rather than a gradient: above roughly 0.15 the loop compounds across rounds, and below it the loop is harmful. We report Π as the primary metric and treat the gate as a precondition for running the pipeline at all.

3 Algorithms

This section describes the objectives that produce the embedding, the statistic that scores it, and the rule that turns that statistic into a pool. The three are usually treated as separate literatures; a contribution of this work is that the middle one — the Neyman–Pearson density ratio — is the same object in all three roles, which is what makes the last one derivable rather than tuned.

3.1 Objectives

We work within a design cube spanning {instance, supervised} positives $\times$ {cosine, bilinear, distance} critics $\times$ {softmax, NPLM} estimators $\times$ {none, SIGReg, class-wise SIGReg} marginals, which contains SimCLR, SupCon, LeJEPA/VISReg and the density-ratio objectives as corners. Of the ten arms we ran, three earn a place for discovery.

SupCon + SIGReg (strong). Supervised positives, cosine critic, softmax estimator, and a sliced isotropic-Gaussian regulariser at weight $\lambda = 5$. This is the discovery workhorse: it produces the highest pool purities we measure. Two points of care. First, λ matters and is not a formality — the same objective at the library default $\lambda = 1$ is a different algorithm and its numbers must not be pooled with these (Appendix A). Second, we make no claim that SIGReg helps *because* it renders distance a log-likelihood; that mechanism is tested and rejected in Section 6. SIGReg earns its place empirically, and structurally: under an isotropic marginal Euclidean distance is Mahalanobis distance, which is what licenses the unit-variance clustering criterion used downstream.

Supervised distance-NPLM. Supervised positives, distance critic, and the NPLM estimator

$$\mathcal{L}[g] \;=\; \mathbb{E}_{\mathcal{R}}[e^{g} - 1] \;-\; \mathbb{E}_{+}[g]\,, \tag{1}$$

whose minimiser is the *absolute* log density ratio, fixed by the calibration $\mathbb{E}_{\mathcal{R}}[e^{g^\star}] = 1$ with no free additive constant. Unlike a softmax-normalised critic, this makes a single embedded distance a statement about relative likelihood rather than a statement only about rank. It is the strongest arm in the low-class-count regime and the most robust across pretraining backbones.

SupCon (plain). Supervised positives, cosine critic, softmax estimator, no marginal. Retained as the baseline and as the parent for the residual construction. It is the best space in the paper by linear probe and by semantic agreement, and one of the weakest by every calibration measure — which is the dissociation this paper is organised around, in a single arm.

We drop plain SimCLR, LeJEPA/SIGReg-SSL, instance NPLM-bilinear and instance NPLM-distance to a single ablation row: none reaches the Pareto front for discovery, and the bilinear critic in particular carries roughly three times the seed variance of the distance critic.

3.2 The Neyman–Pearson statistic, in three roles

Let $\mathcal{R}$ be the reference and $\mathcal{D}$ the corpus, with $w = N_{\mathcal{D}}/N_{\mathcal{R}}$. The empirical NP objective

$$\mathcal{L}[f] \;=\; w \sum_{x \in \mathcal{R}} \Big(e^{f(x)} - 1\Big) \;-\; \sum_{x \in \mathcal{D}} f(x) \tag{2}$$

has minimiser $f^\star(x) = \log\big(p_{\mathcal{D}}(x)/p_{\mathcal{R}}(x)\big)$, obtained by setting the functional derivative $p_{\mathcal{R}} e^{f} - p_{\mathcal{D}}$ to zero, and therefore satisfies

$$\mathbb{E}_{\mathcal{R}}\Big[e^{f^\star}\Big] = \int p_{\mathcal{R}} \cdot \frac{p_{\mathcal{D}}}{p_{\mathcal{R}}} = 1. \tag{3}$$

Equation 2 is Equation 1 with pairs replaced by events. The same functional appears as a contrastive *loss* over pairs, as a two-sample *test statistic* $t_{NP} = -2\mathcal{L}$ distributed as a Wilks log-likelihood ratio, and — below — as a *pool scorer*. We parameterise f as a small kernel ensemble with learnable centres and amplitudes and anneal the bandwidth from the median pairwise distance downwards; a fixed bandwidth is calibrated only for one embedding scale and goes blind when the geometry is rescaled.

Equation 3 is also a free diagnostic, and we report it. A fitted critic need not satisfy it, and a value far from 1 means the numbers built from that fit are unreliable. Measured on trained spaces it is 0.997 to 1.019 in-sample; out-of-sample it ranges to 2.7, reflecting mild overfitting of the reference subsample.

Why the ratio and not the distance. Distance-based pooling asks which points are far from everything. The density ratio asks which points are over-represented relative to what the reference explains. These come apart, and the difference is decisive: a novel class is not a scatter of far-flung outliers but a *dense clump* sitting where the reference has no mass. On a 100-class benchmark the extreme distance tail is owned by background outliers, and tightening the distance threshold makes purity *worse* ($0.036 \to 0.002$), whereas tightening the ratio threshold makes it better ($0.232 \to 0.358$). Same space, same rate, opposite response.

3.3 Choosing the pool: an algebra and a rule

The algebra. Let $q = |\mathcal{P}|/N$ and let $\varepsilon_s, \varepsilon_b$ be the fractions of novel and background points retained. Then $q = \varepsilon_s b + \varepsilon_b(1-b)$ and

$$\Pi \;=\; \frac{\varepsilon_s\, b}{q}, \qquad E \;\equiv\; \frac{\Pi}{b} \;=\; \frac{\varepsilon_s}{q}, \qquad \Pi \;\le\; \min\Big(1,\; \frac{b}{q}\Big). \tag{4}$$

Three consequences. The enrichment is exactly the lift at operating point q. Purity increases monotonically as q shrinks, so there is no interior optimum and targeting a purity *level* is the wrong objective. And the ceiling $\min(1, b/q)$ is a property of the cut alone: at $q = 0.05$ and $b = 0.01$ no scorer, however good, can exceed $\Pi = 0.20$. A threshold chosen without reference to b can therefore cap purity near the gate independently of the representation — and the widely used "95th percentile of the reference" is such a threshold.

The rule. Purity cannot be used to choose q, since computing it requires the novel labels. The critic supplies the missing quantity. Under the contamination model $p_{\mathcal{D}} = (1-b)p_{\mathcal{R}} + b\, p_S$, with $r = e^f$, the posterior that a corpus point is novel is $1-(1-b)/r$; dropping the unknown b and taking the positive part gives the label-free weight

$$w(x) \;=\; \left[1 - \frac{1}{r(x)}\right]_+, \qquad \mathbb{E}_{\mathcal{D}}[w] \;=\; \mathrm{TV}(p_{\mathcal{D}}, p_{\mathcal{R}}) \;=\; b\,\mathrm{TV}(p_S, p_{\mathcal{R}}) \;\le\; b, \tag{5}$$

so $\sum_{x\in S} w(x)$ estimates how many novel points a subset S contains, exactly when the novel class is disjoint from the reference and biased *downward* under overlap — a safe direction, since it tightens the cut. Since purity rises as q falls, the binding constraint is not purity but whether a cluster survives, and that is what we target: ordering the corpus by f and writing $\widehat{N}(k)$ for the null-corrected sum of w over the top k points,

$$k^\star \;=\; \min\big\{\, k : \widehat{N}(k) \ge n_{\min} \,\big\}, \qquad q^\star = k^\star/N, \qquad k_{\max} = \Big\lfloor \widehat{N}(N)\,/\,n_{\min} \Big\rfloor. \tag{6}$$

The same weights bound the clustering search range, replacing a bound that in the original pipeline was set from the number of held-out classes — a quantity an open-world method does not have.

The null correction, which is not optional. Equation 5 is a positive part, so *any* spread in a fitted f manufactures apparent novelty. On pure-noise scores with no novel class at all, the raw sum returns 665 "novel" points out of 8000 — enough for the rule to fire on nothing and hand the loop a pool of pure background. Subtracting the mean weight of the reference does not repair this, because top-ranked null points carry far above-average weight and their tail survives subtraction. We instead subtract a *rank-matched* null: under the null hypothesis the top q of the corpus is distributed as the top q of the reference, so $\widehat{N}(k) = \sum_{i \le k} w(x_{(i)}) - k \cdot \overline{w}_{\mathcal{R}}^{(q)}$, with $\overline{w}_{\mathcal{R}}^{(q)}$ the mean weight of the top-q fraction of the reference. This cancels to zero under the null while preserving a real excess. When $\widehat{N}(N) < n_{\min}$ the rule declines to pool and reports that it has declined; that refusal is an output of the method, and we report it as one.

Constants. $n_{\min}$ and the clipping range on q are set once, globally, from a sweep on held-out embeddings rather than inherited: purity falls monotonically as $n_{\min}$ grows, so it is set as small as the clustering will bear.

4 Constructions

Both constructions in this section are defined *relative to an anchor set*, which is what lets them compose with the discovery loop: discovery enlarges and corrects the anchors, so a construction defined against them is a different construction after discovery than before. That composition is the part of this work with no counterpart we are aware of, and it is also the part most easily overstated, so we measure it as an interaction rather than a delta.

4.1 The residual decomposition

Let μ_y be the anchor for class y and z an embedding. The residual $r = z - \mu_y$ is what the anchor set fails to explain about z. We train a second encoder on r under the same objective family as the parent and either use it alone (*residual*) or concatenate it with the parent (*concat*).

The motivation is that a supervised objective is free to discard any variation that does not separate the classes it was given, and novelty is precisely variation of that kind. The residual target makes that discarded structure the only thing the child can see.

On its own, the residual buys detection at no accuracy cost. On the from-scratch CIFAR-100 lineage, concatenating a residual onto a supervised contrastive parent moves Euclidean novelty AUC from 0.233 to 0.521 while the linear probe moves by $+0.0015$, a gain two orders of magnitude smaller than the detection gain and of the same order as the probe's own seed spread ($\sigma = 0.0008$). This is the pattern across cells: the pure residual usually *loses* probe accuracy (four of six strong-parent cells, -0.037 to -0.077), and the concatenation recovers it. We therefore report concat as the construction and the bare residual as a diagnostic.

4.2 Composing with discovery: a 2×2

Because the residual is defined against the anchors, it can be built before or after the discovery loop. Writing A for the parent alone, B for the parent after discovery, C for the construction before discovery, and D for both, the quantity of interest is not $D - A$ but the interaction

$$\Delta_{\text{int}} \;=\; (D - C) - (B - A), \tag{7}$$

which is zero for an additive stack and positive only when the two ingredients do more together than apart. We report it for per-event power at a nominal false-positive rate $\alpha = 0.05$, because that metric is bounded below by α: a parent at $A \approx \alpha$ has *no* ability to flag an individual novel sample, so a positive interaction there is a threshold crossing rather than an improvement.

The interaction is real, and it is narrow. On DTD with a DINO backbone, the residual construction gives $A = 0.040$, $B = 0.068$, $C = 0.023$, $D = 0.253$. Discovery alone is worth $+0.028$; the construction alone is worth -0.018 — it actively hurts — and composed they reach $6\times$ nominal, an interaction of $+0.203$. The concatenation behaves the same way ($+0.163$). Neither ingredient does anything alone. This is the strongest evidence we have that discovery and representation construction are not independent improvements.

Two apparent replications are not replications. The same protocol on galaxy10 produces the two largest raw gains in the study, and the decomposition shows that neither is composition. With a DINO backbone the residual takes per-event power from 0.021 to 0.299, but $C = 0.281$ already: the construction does this *by itself*, and the interaction is $+0.003$. With a LeJEPA backbone the gain from 0.007 to 0.206 is almost entirely the discovery main effect ($+0.190$), with an interaction of $+0.008$ — and it occurs in a cell whose round-2 pool purity is exactly zero. That last case is worth stating plainly, because we had written it in advance as a falsifier: if the composition helps where purity is zero, then the mechanism we proposed for it — that discovery supplies a richer and *correct* anchor set, against which the residual becomes informative — cannot be what is happening there.

What we therefore claim. Across five cells, two constructions, and two variants, 12 of 20 interactions are positive, but most are small and the threshold crossings driven by interaction rather than by a main effect occur in a single cell with a single construction. We do not claim superadditivity as a general property. The claim the data supports is weaker and still useful: *discovery and the residual are substitutes more often than complements.* Whichever of the two a cell is missing, one of them supplies it — galaxy10/LeJEPA needed discovery, galaxy10/DINO needed the residual, DTD/DINO needed both — and a practitioner who runs only one of them will, on most cells, capture most of the available gain. Every number in this subsection is a single seed; the interaction is a difference of differences and inherits the variance of both its terms, so we treat it as a finding to be replicated rather than a headline.

4.3 Frozen-space discovery

5 Experiments

6 When it does not work

A Objective naming